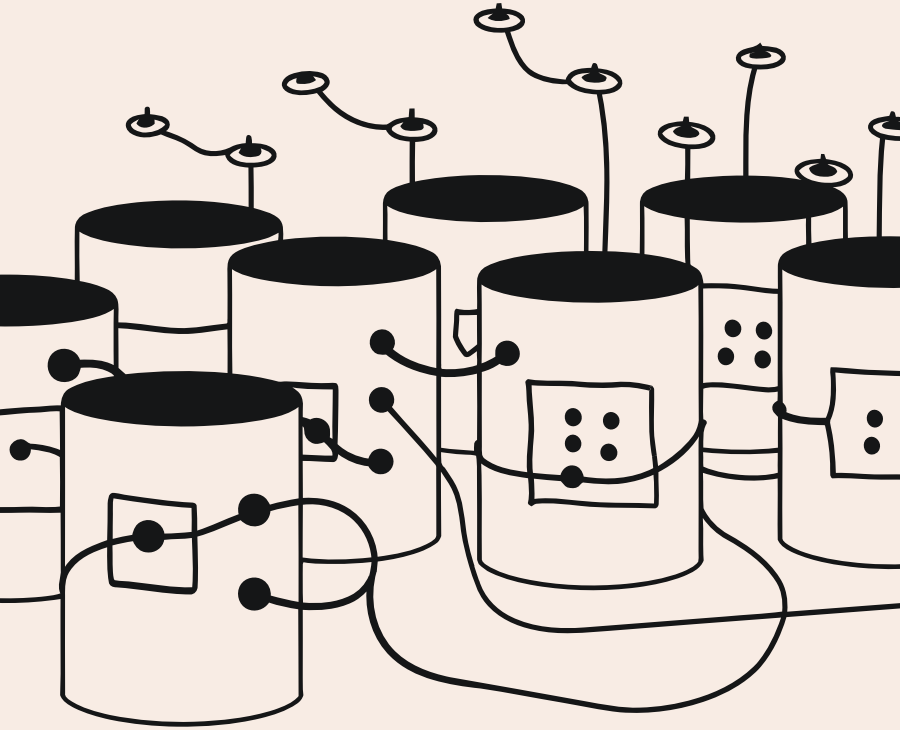




Conversational Knowledge Graph Extraction

v1 - 21/05/2026

Research Questions

❓ Does **conversational graph extraction** produce a better schema than a single prompt?

Secondary Questions

❓ Does it generalize, or overfit user conversation?

Positioning

No prior work studies KG schema from **free dialogue with a non-expert** and no upfront goal.

Evaluation

Baseline: traditional KG extraction without LLMs

Split	validation	test
one-shot	✓ conversational	✓ Generalization
conversational	✓ conversational	✓ Generalization

Ground Truth

Label a subset of files to check quality and generalization.

LLM Judge

Use an LLM to scale annotations

❓ How to asses LLM quality?

Downstream Evaluation

Test graph utility on person relationships

Metrics

- Precision and recall
- Generalization gap
- Schema coverage
 - new types of relations not accounted for





Related Work & Toolstack

Key Papers

- **Extract, Define, Canonicalize** – LLM framework for KG construction (arXiv:2404.03868)
- **AutoSchemaKG** – Autonomous schema induction at web scale (arXiv:2505.23628)
- **SCHEMA-MINER** – Human-in-the-loop workflow closest to this work (arXiv:2504.00752)
- **LLM as Ontology Copilot** – Practical, prompt-sensitive approach (arXiv:2403.08345)

Tool Decisions

spaCy

NLP baseline

SBERT

Schema similarity

NetworkX

KG analysis

NumPy / Matplotlib

Analysis and charts

Dataset: The Epstein Files



About the Data

Public court and miscellaneous documents from the Jeffrey Epstein case.

Considerations

Data Quality

OCR and PDF extraction can add noise; quality checks are needed.

Ethics & Privacy

Files are public and already censored by US authorities.